Sorting factory lesson plan

22 October 2026 · Four hours · First semester · No coding or robotics prerequisites

Work in pairs. One partner predicts, one types; swap often. Every task follows predict, change one thing, run, check. Use the prepared local Jupyter notebook; no software installation during class. The complete lesson also works with offline controlled examples. Red goes left, blue goes right, everything else goes to reject.

Elapsed minutes	Duration	Activity	Evidence of learning
00–25	25 min	Introduction and notebook demonstration	Identify sensor, decision, actuator and feedback
25–35	10 min	Blockly Maze levels 1–3	Predict a sequence and fix one wrong step; levels 4–5 optional
35–65	30 min	A Python names, repeats and decisions	Change one value; distinguish = from ==
65–85	20 min	B Teachable Machine perception	Compare red, blue and background; test changed lighting or background
85–95	10 min	Break	Save and assign rotation groups
95–140	45 min	C Three stations of 15 min	Robot and twin; camera and conveyor; virtual robot
140–190	50 min	D Complete sorting factory	Trace stop, observe, decide, sort and compare routes and totals
190–225	35 min	E Challenges	Two predictions, one change per experiment, recorded evidence
225–240	15 min	Demos and reflection	Show one change, one result and one real-world limitation

Total 240 minutes including transitions. At each rotation use 2 minutes to explain, 9 to try, 3 to explain a result and 1 to save and rotate. Three groups rotate in opposite starting positions: group 1 robot → camera → virtual; group 2 camera → virtual → robot; group 3 virtual → robot → camera.

The robot station uses Robot_Candy_Factory_Live_SO101.ipynb. Its default data are invented DEMO examples. A separately launched LIVE mode reads existing ROS topics only; the instructor controls physical enable, mode and stop using the RViz operations panel. The virtual station uses the short command sequence in the student notebook. The camera station combines observation uncertainty with paper conveyor sequencing; only the instructor demonstrates qualified hardware.

For the 190–225 minute challenge, pairs can choose the offline Blockly factory or the notebook challenges. Start with simulation on every laptop. At the robot station, the instructor can demonstrate recording one fixed-jaw teleoperation stroke and replaying it through the existing robot controller. This replaces time within the challenge; it adds no extra lesson slot. A physical dispense/pick/place sequence additionally needs calibrated grip/release and qualified strokes. See the instructor Blockly guide.

Use the worksheet to record answers; use the student notebook for explanations. External sites and their offline alternatives are in workshop-resources.md. The handbook reproduces the notebook text and code; build_notebooks.py is its source generator. Integrated physical sorting remains subject to the station's commissioning and acceptance evidence, independently of the notebook tests.